

Problem Set 9:

Submission: 12:00, Friday, June 12, 2026; Solutions: Friday, June 26, 2026

Problem 1 (Central exercise) An Oscillating Point Charge on a Spring

A point charge with charge q and mass m is attached to the end of a spring with spring constant k . The charge is displaced in the z -direction by an amount α from its equilibrium position and then released, so that it oscillates. Assume that the resulting motion is nonrelativistic, $v \ll c$.

- (1.a) Assume that the charge oscillates harmonically with amplitude α . What electromagnetic potentials ϕ and $\mathbf{A}$ are obtained up to order $1/r$ far from the charge and to leading order in v/c ?
- (1.b) What is the radiated power?
- (1.c) Due to the radiation of electromagnetic energy, the maximum oscillation amplitude α will in fact slowly decrease with time. Determine $\alpha(t)$.

Problem 2 Radiated Power of a Rotating Square of Charges

A square with side length d lies in the plane $z = 0$, with the center of the square located at the origin, i.e. at $x = y = z = 0$. Charges are placed at the corners of the square in two different configurations.

The square rotates with angular velocity Ω about the z -axis, where

$$\Omega d \ll c \quad (1)$$

is assumed to hold.

The four vertices of the square are

$$\left(\frac{d}{2}, \frac{d}{2}, 0\right), \quad \left(-\frac{d}{2}, \frac{d}{2}, 0\right), \quad \left(-\frac{d}{2}, -\frac{d}{2}, 0\right), \quad \left(\frac{d}{2}, -\frac{d}{2}, 0\right). \quad (2)$$

Consider the following two charge configurations:

- (i) The charges are given by

$$q \text{ at } \left(-\frac{d}{2}, \frac{d}{2}, 0\right), \quad q \text{ at } \left(\frac{d}{2}, \frac{d}{2}, 0\right), \quad (3)$$

and

$$-q \text{ at } \left(-\frac{d}{2}, -\frac{d}{2}, 0\right), \quad -q \text{ at } \left(\frac{d}{2}, -\frac{d}{2}, 0\right). \quad (4)$$

(ii) The charges are given by

$$q \text{ at } \left(-\frac{d}{2}, \frac{d}{2}, 0\right), \quad q \text{ at } \left(\frac{d}{2}, -\frac{d}{2}, 0\right), \quad (5)$$

and

$$-q \text{ at } \left(\frac{d}{2}, \frac{d}{2}, 0\right), \quad -q \text{ at } \left(-\frac{d}{2}, -\frac{d}{2}, 0\right). \quad (6)$$

In each case, the radiated power P scales with q , d , and Ω as

$$P \propto q^{n_1} d^{n_2} \Omega^{n_3} \quad (7)$$

for certain integers n_1 , n_2 , and n_3 . Determine n_1 , n_2 , and n_3 for configuration (i) and for configuration (ii).

Problem 3 Expansion of a Plane Wave in Spherical Harmonics

Show that e^{ikz} can be expanded in spherical harmonics as

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^{\ell} \sqrt{4\pi(2\ell+1)} j_{\ell}(kr) Y_{\ell 0}(\theta, \varphi) = \sum_{\ell=0}^{\infty} i^{\ell} (2\ell+1) j_{\ell}(kr) P_{\ell}(\cos \theta). \quad (8)$$

Note that this formula allows one to write a plane-wave solution of the scalar wave equation in terms of spherical waves.

Hint: First show that

$$e^{-ikz} = \lim_{\lambda \rightarrow \infty} \lambda e^{-ik\lambda} \frac{e^{ik|\mathbf{x} - \lambda \hat{\mathbf{z}}|}}{|\mathbf{x} - \lambda \hat{\mathbf{z}}|}. \quad (9)$$

Then use the expansion formula

$$G^H(\mathbf{x}, \mathbf{x}') = \frac{e^{i\omega|\mathbf{x} - \mathbf{x}'|/c}}{4\pi|\mathbf{x} - \mathbf{x}'|} = \begin{cases} \sum_{\ell, m} \frac{i\omega}{c} j_{\ell}\left(\frac{\omega r}{c}\right) h_{\ell}^{(1)}\left(\frac{\omega r'}{c}\right) Y_{\ell m}^*(\theta', \varphi') Y_{\ell m}(\theta, \varphi), & r < r', \\ \sum_{\ell, m} \frac{i\omega}{c} h_{\ell}^{(1)}\left(\frac{\omega r}{c}\right) j_{\ell}\left(\frac{\omega r'}{c}\right) Y_{\ell m}^*(\theta', \varphi') Y_{\ell m}(\theta, \varphi), & r > r', \end{cases} \quad (10)$$

as well as the asymptotic form

$$h_{\ell}^{(1)}(x) = (-i)^{\ell+1} \frac{e^{ix}}{x} + O\left(\frac{1}{x^2}\right), \quad (11)$$

of $h_{\ell}^{(1)}(x)$ and

$$Y_{\ell m}(0, 0) = \sqrt{\frac{2\ell+1}{4\pi}}. \quad (12)$$